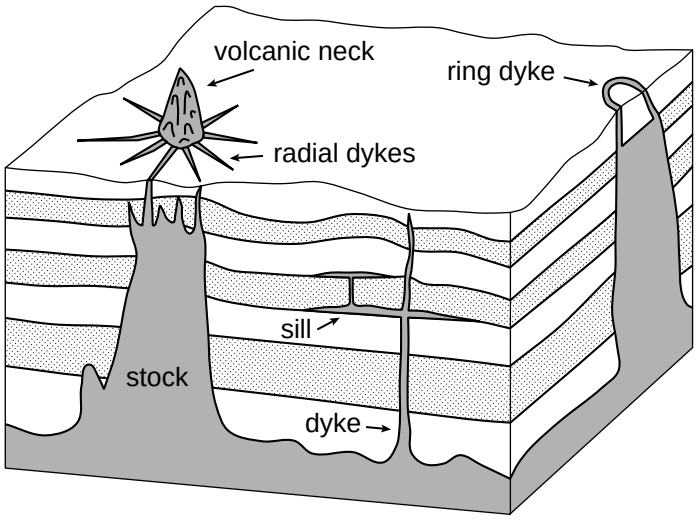
THE GEOMETRY OF INTRUSIVE IGNEOUS BODIES (1)

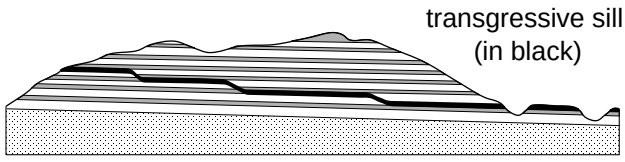
Igneous bodies may be ***extrusive*** onto the Earth's surface or ***intrusive*** into the rocks forming the Earth's crust. The geometry of extrusive igneous units (*e.g.*, lava flows or pyroclastic deposits) can be considered in the same way as that of sedimentary units, although laterally their thickness may be much more variable. Consequently, they will not be considered further.

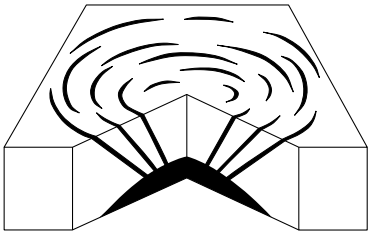
• The generic term for an intrusive igneous body, irrespective of its size, shape, or composition, is ***pluton***.
• Plutons are referred to as ***composite*** if they bear evidence of more than one phase of emplacement of materials of different composition, or as ***multiple*** if they bear evidence of more than one phase of emplacement of materials of the same composition.
• From the geological map interpretation perspective, the geometry of plutons is considered in terms of
(a) the shape of the pluton itself (*i.e.*, whether it is ***parallel-sided*** or of ***more complex shape***), and
(b) the orientation of its boundaries with respect to the layering in the units (*country rock*) it intrudes (*i.e.*, whether these cut across that layering, in which case the pluton is referred to as ***discordant***, or whether these are predominantly parallel with it, in which case the pluton is referred to as ***concordant***).

Parallel-sided intrusives

Parallel-sided intrusions are planar or arcuate in form and are of approximately constant thickness over distances much greater than their thickness.
• ***Sill*****:** a concordant, sheet-like intrusion, that is, one which is intruded parallel to the layering in the country rock. Sills may be restricted to a single surface but those which have large geographical extent commonly step across surfaces, and where this stepping is progressively upward to higher structural levels, the sill is described as *transgressive*. The 'original' sill is the Great Whin Sill which underlies much of Northumberland and County Durham.
• ***Dyke*****:** a discordant, sheet-like intrusion, that is, one which cuts across the layering in the country rock. Dykes commonly occur in large *swarms* which may be ***parallel*** or ***radial*** in nature.
• ***Cone-sheet*****:** a dyke which has the form of part of a conical surface (*i.e.*, which has an arcuate outcrop pattern and inwardly dipping contacts). Typically cone-sheets occur in swarms about a common centre.
• ***Ring dyke*****:** dykes with an approximately circular or part circular outcrop pattern, which have vertical to steeply outwardly dipping contacts.







THE GEOMETRY OF INTRUSIVE IGNEOUS BODIES (2)

Intrusives of more complex shape

• ***Batholith*:** a discordant, composite intrusion, comprising numerous plutons, which is over 100 km^2 in outcrop area, and which has no observable bottom. Batholiths are typically found in the cores of orogenic belts, elongate parallel to them. they are generally composed of thick, horizontal, sheet-like or lobate bodies which extend from a central funnel-shapped region.

• ***Stock*:** commonly used to refer to a discordant, irregularly-shaped pluton which is smaller in size than a batholith (*i.e.*, less than 100 km^2 in outcrop area). It may also be used more strictly to refer to an intrusion with an approximately circular or elliptical outcrop pattern, which is less than 20 km^2 in size, and which has contacts that are steep or gently dipping.

• ***Boss*:** broadly synonymous to 'stock', being a discordant pluton with a circular outcrop, which is less than 25 km^2 in size, and which has steep contacts.

• ***Cupola*:** a small dome-like protuberance projecting from the main body of a larger igneous intrusion but isolated from it in map view.

• ***Plug*:** *sensu stricto* − the approximately circular (in outcrop) and vertical feed channel of a volcano which has been infilled with solidified magma and/or pyroclastic material; when exposed after erosion of the surrounding volcanics, it may be referred to as a *volcanic neck*; a French synonym, especially as applied to the volcanic necks of the Auvergne district of Massif Central, is *puy*. *Sensu lato* − any rather small, approximately vertical cylindrical unit of igneous rock; an old-fashioned synonym is *bysmalith*.

• ***Diatreme*:** a deep (typically 1-2 km) carrot-shaped (*i.e.*, with circular to elliptical cross-section, vertical axis, and steeply inwardly dipping sides) volcanic vent generally infilled with breccia.

• ***Laccolith*:** a concordant, dome-like intrusion which arches the overlying units and has an approximately flat floor. Laccoliths are commonly circular or elliptical in outcrop pattern. A cedar-tree laccolith is one where a series of laccoliths are stacked one on top of the other. Laccoliths were first described in the Henry Mountains, Utah.

• ***Lopolith*:** a concordant, saucer-shaped intrusion which is concave-upwards. Small lopoliths may arise from intrusion into synformally folded country rocks. However, the name is also applied to large basin-shaped intrusions such as the Bushveld complex in southern Africa where control by folding is less clear.

• ***Phacolith*:** a concordant, concavo-convex intrusion. Phacoliths are generally considered to arise from intrusion into folded country rocks, or from intrusion during folding. They may occur in stacked sets, and in this form are sometimes called *saddle reefs*.

